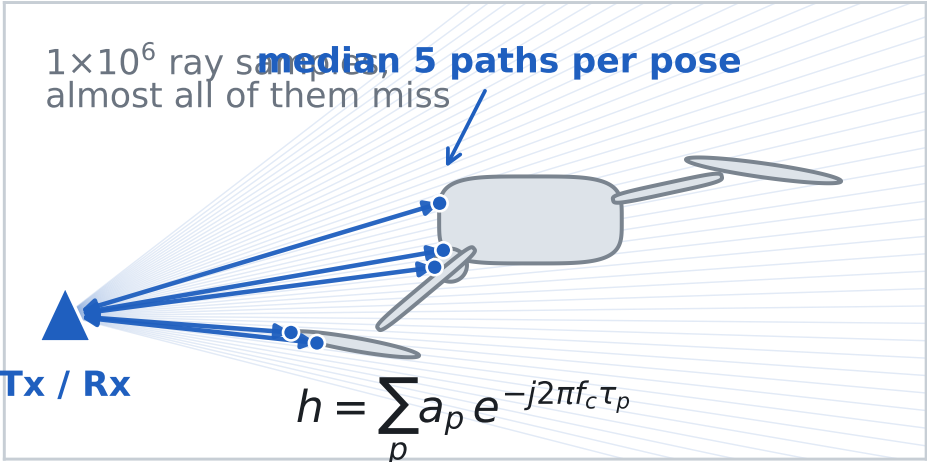
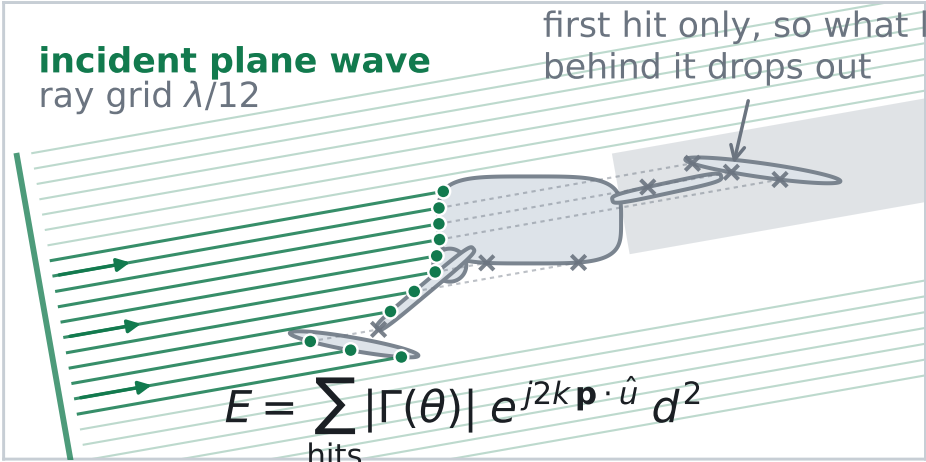


What each of the three engines actually computes

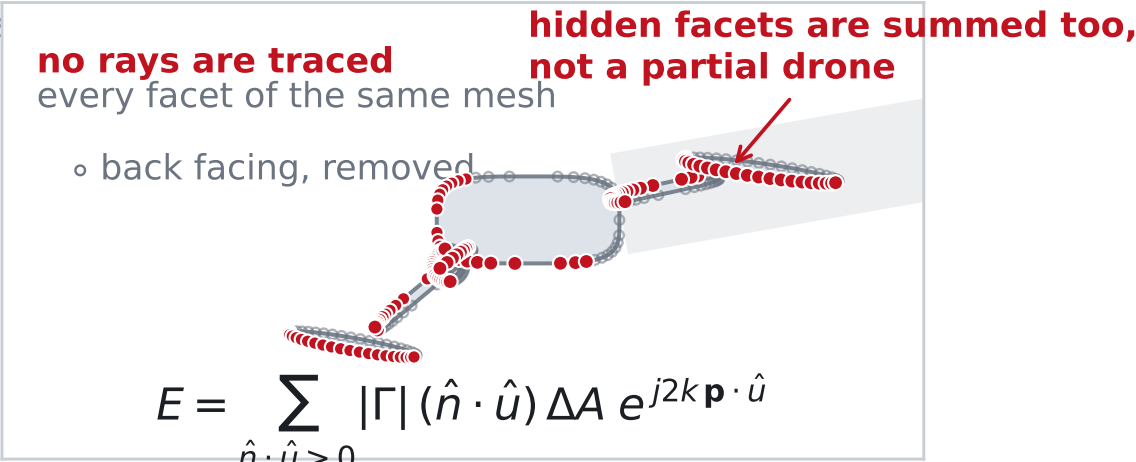
Sionna PathSolver



Ours (SBR+PO, default)



Ours w/o occlusion (control)



FIRED ray samples fired from the Tx

SUMMED the paths that reach the Rx again

LEFT OUT the surface integral over the target

FIRED plane wave on a $\lambda/12$ ray grid, 11k to 16k rays

SUMMED the first hit of every ray

LEFT OUT back facing and hidden facets

FIRED nothing is fired, no ray tracing

SUMMED every facet of the same mesh, body and rotors

LEFT OUT back facing facets only

- Schematic of the three kernels, not a computed result. The airframe is a readable silhouette rather than the simulated mesh.
- The two right hand columns differ in more than occlusion. They also differ in discretisation (a $\lambda/12$ ray grid against surface point clouds at $\lambda/11$ on the blades and $\lambda/6$ on the frame) and in the angle dependent $|\Gamma(\theta)|$, which is on in the middle column and off in the control. Their gap is therefore not the price of occlusion alone.
- The left column is lit by a point source at 3 m while the other two use an ideal plane wave. The middle column also lets rays through the plastic shell so that the battery and the board inside are counted.